Name: _____ Period: _____

IN CLASS LAB 2

Mineral identification

How can minerals that look similar be correctly identified using their physical properties?

LEARNING INTENTION

We are learning to identify minerals using evidence and explain how structure affects their properties.

SUCCESS CRITERIA

- I can test and record a mineral's physical properties, including luster, streak, breakage, and hardness.
- I can use multiple physical properties as evidence to correctly identify an unknown mineral.
- I can explain how a mineral's internal structure and bonding affect its physical properties.

Procedure: Observe each mineral sample using a hand lens. Record visible properties: luster, streak, breakage, hardness and color by following the steps below:

PHYSICAL PROPERTY	HOW TO IDENTIFY
LUSTER	1. Hold the mineral under a light source. 2. Slowly tilt the mineral and observe how light reflects off its surface. 3. Decide whether the mineral has: Metallic luster (*does it shine like a metal?*) or Nonmetallic luster
STREAK	1. Hold the mineral firmly and rub it across the streak plate. 2. Observe the color of the mark left on the plate, not the color of the mineral itself 3. If no streak is produced, record "no streak". NOTE: Do not press excessively hard. The goal is to produce powder, not to break the sample.
BREAKAGE	1. Examine broken surfaces of the mineral using a hand lens. 2. Look for flat, smooth surfaces that repeat in the same direction. 3. Decide whether the mineral shows cleavage or fracture. A mineral has cleavage if it: ● Breaks along flat, smooth surfaces ● Breaks in repeating directions ● Produces pieces with similar shapes (Flat sheets, Box-like shapes, Step-like surfaces)
HARDNESS	Glass has a hardness of approximately 5.5 on the Mohs scale. Instead of assigning an exact hardness number, you will determine whether each mineral is harder than glass or softer than glass. 1. Attempt to scratch the glass plate with the mineral. 2. Observe whether a visible scratch is produced. 3. Record your observation as: Harder than glass (glass is scratched) or Softer than glass (glass is not scratched). NOTE: Do not press excessively hard. A true scratch should remain after wiping the glass clean.

***Use ESSRT pages 16 and 17 to name your mineral**

#	Color	Streak	Breakage	Luster	Hardness	Name of Mineral
			☐ Cleavage ☐ Fracture	☐ Metallic ☐ Nonmetallic	☐ Harder than glass ☐ Softer than glass	
			☐ Cleavage ☐ Fracture	☐ Metallic ☐ Nonmetallic	☐ Harder than glass ☐ Softer than glass	
			☐ Cleavage ☐ Fracture	☐ Metallic ☐ Nonmetallic	☐ Harder than glass ☐ Softer than glass	
			☐ Cleavage ☐ Fracture	☐ Metallic ☐ Nonmetallic	☐ Harder than glass ☐ Softer than glass	
			☐ Cleavage ☐ Fracture	☐ Metallic ☐ Nonmetallic	☐ Harder than glass ☐ Softer than glass	

Follow Up Questions

1. Are all mineral properties equally useful for identification? Explain.

2. Why is color alone often an unreliable property for identifying minerals?

3. Two minerals may look very similar but still be different minerals. Which properties would help distinguish them?

4. Would any mineral properties change if the sample were larger? Explain your answer.

Claim–Evidence–Reasoning (CER)

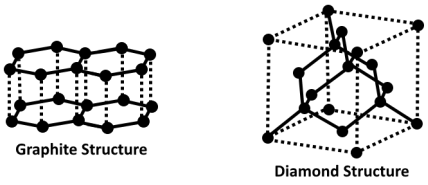
Graphite and diamond are both minerals made entirely of the element carbon. Even though they have the same chemical composition, their physical properties are extremely different. This difference is caused by how the carbon atoms are arranged and bonded.

In graphite, carbon atoms are arranged in thin, flat layers. Within each layer, the atoms are strongly bonded, but the bonds between layers are weak. Because the layers can slide past one another easily, graphite is very soft and can be used in pencils. Graphite has a nonmetallic luster and does not scratch glass.

In diamond, carbon atoms are bonded in a rigid, three-dimensional network. Each atom is strongly bonded to multiple neighboring atoms, creating an extremely strong structure. This arrangement makes diamond the hardest known mineral. Diamonds can easily scratch glass and many other materials.

Although graphite and diamond contain the same element, their different atomic structures result in very different physical properties.

Allotropes of carbon



Question: Graphite and diamond are both made of carbon. Why do these two minerals have such different physical properties?

Make a claim explaining why diamond is much harder than graphite, even though both minerals are made of carbon.

Claim: *(Diamond is harder than graphite because)*

Use two pieces of evidence from the reading to support your claim.

Evidence 1:

Evidence 2:

Use complete sentences to explain how the internal arrangement of atoms in a mineral influences its physical properties.
In your explanation, describe how:

- The way atoms are arranged within the mineral
- The strength of the bonds between those atoms

Reasoning:

Reflection Questions:

1. If two minerals had the same color, luster, and streak, what additional property would you test next? Why?

2. Why do scientists use multiple tests rather than a single property to identify minerals?

3. Why is **hardness testing** considered one of the most useful tools for mineral identification?

Cluster Set

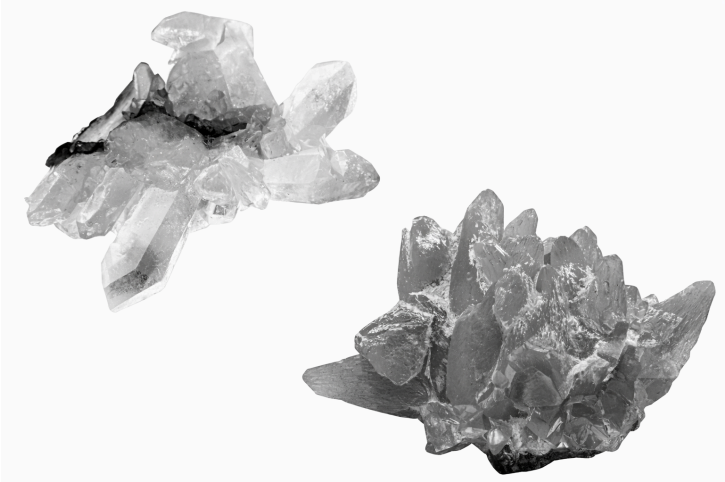
Identifying Look-Alike Minerals

Two clear, crystal-shaped minerals are placed side by side. At first glance, they appear nearly identical. Both are transparent, glassy, and form well-defined crystals. A person with no scientific training might reasonably assume the minerals are the same.

However, when these minerals are tested for use in making glass, only one of them produces glass that resists scratching. The other results in glass that scratches easily and wears down over time. Despite their similar appearance, the two minerals behave very differently when used in real-world materials.

This raises an important question for Earth scientists and materials engineers: How can scientists determine which mineral is which if appearance alone is misleading?

To answer this question, scientists must rely on measurable physical properties that reflect a mineral's internal structure rather than how it looks on the outside.



1. Which claim best explains how scientists can determine which mineral is which?
   (1) Scientists compare the color and shape of the minerals.
   (2) Scientists use physical properties that reflect internal structure.
   (3) Scientists measure the mass of each mineral sample.
   (4) Scientists identify the mineral that looks the shiniest.

2. Which piece of evidence would be most useful for determining which mineral can be used to make scratch-resistant glass?

   Choose two.
   - ☐ Whether the mineral is harder or softer than glass
   - ☐ The color of the mineral
   - ☐ Whether the mineral shows cleavage or fracture
   - ☐ The size of the mineral sample

3. Explain why the evidence you selected in Question 2 is more reliable than the other options. Use evidence-based reasoning in your answer.

4. A mineral sample is clear and glassy. It scratches glass and breaks with curved surfaces rather than flat planes.

    a. Is this mineral harder or softer than glass? _____

    b. Which property helped you determine this? _____

    c. Why does this property matter more than appearance?

    d. How does a mineral's internal structure affect its ability to resist scratching?